

Analysis of Healthcare Data: Language, Race, and Ethnicity

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About the dataset

The dataset used in this project is titled “**Data for healthcare analyst.xlsx**” and contains comprehensive demographic and screening information for **6,597 healthcare members**. It serves as the foundation for exploratory data analysis, predictive modeling, and interactive dashboard development aimed at uncovering disparities in healthcare screening participation.

□ Dataset Overview

Column Name	Description
Member ID Encrypted	Unique identifier for each member (anonymized for privacy)
EBM Numerator	Indicates whether the member completed the recommended screening (1 = Yes)
EBM Denominator	Indicates eligibility for screening (1 = Eligible)
ORIG_PREFERRED_LANGUAGE	Member’s originally preferred language
SRC_LANGUAGE	Language reported by the source system
ORIG_RACE	Original race classification of the member
SRC_RACE	Race classification reported by the source system
Original Member Ethnicity	Member’s original ethnicity
SRC_ETHNICITY	Ethnicity reported by the source system
Zip (5-digit)	Member’s residential ZIP code

★ Key Characteristics

- **Size:** 6,597 rows × 10 columns
- **Type:** Structured tabular data in Excel format
- **Scope:** Demographic profiling, screening eligibility, and participation tracking
- **Use Case:** Enables analysis of screening disparities across race, ethnicity, language, and geography

□ Data Quality Notes

- Presence of discrepancies between original and source-reported demographic fields (e.g., race and ethnicity)
- Missing or “Unknown” entries in ethnicity and race categories
- Requires cleaning and normalization for reliable modeling

🔍 Analytical Potential

This dataset supports:

- **Exploratory Data Analysis (EDA):** Frequency distributions, cross-tabulations, and correlation studies

- **Statistical Testing:** Chi-square and ANOVA to assess demographic influence on screening
- **Predictive Modeling:** Logistic regression, Random Forest, and XGBoost to forecast screening likelihood
- **Visualization:** Power BI dashboards for interactive storytelling and stakeholder engagement

Analysis of Healthcare Data: Language, Race, and Ethnicity

Relevant Inquiries

Q1. Analyze the correlation between the preferred language of members and the EBM Numerator performance across different zip codes.

Grouping and Calculation

- **Data Grouping:** The data was grouped by 'Zip (5-digit)' and 'ORIG_PREFERRED_LANGUAGE' to calculate the average EBM Numerator for each language within each zip code.
- **Average EBM Numerator:** The mean of 'EBM Numerator' was calculated for each group, providing insights into the performance of different language groups across various zip codes.

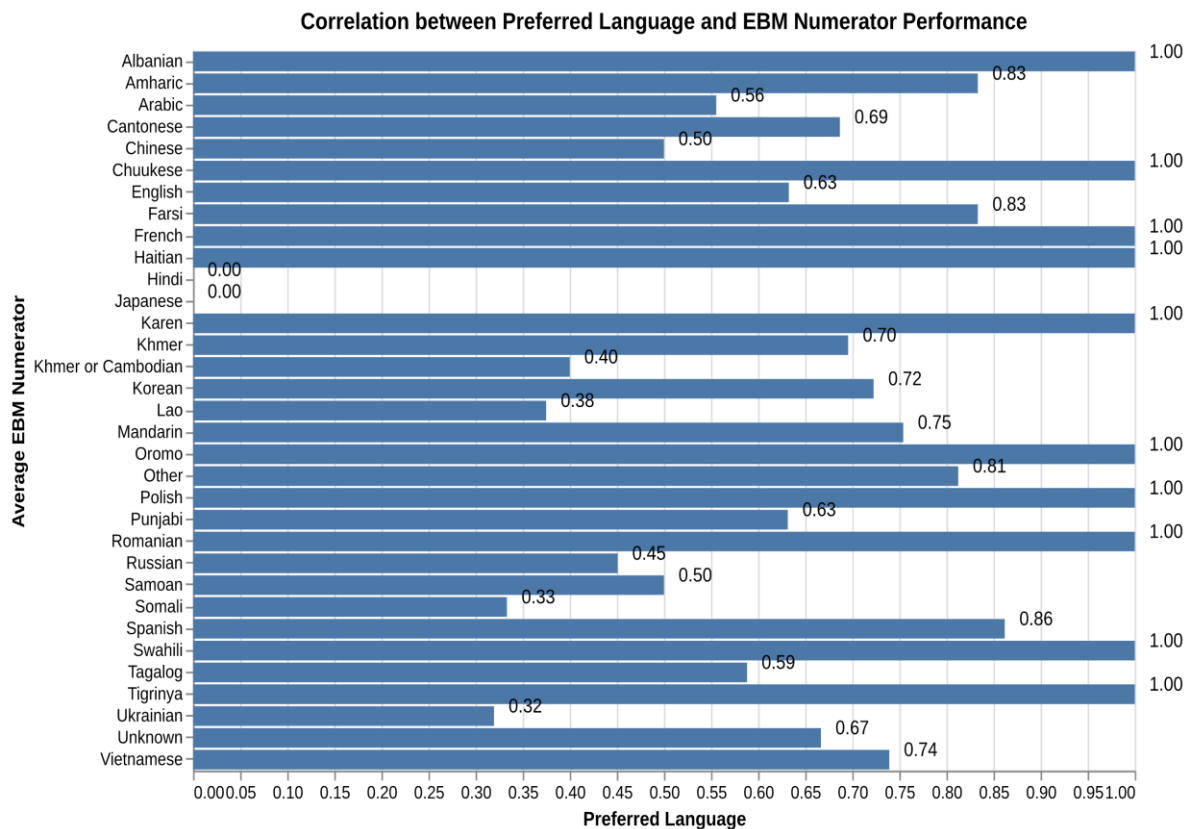
Statistical Overview

- **Average EBM Numerator Statistics:** The 'Average EBM Numerator' ranges from 0 to 1, with a mean of 0.639 and a standard deviation of 0.3649. This indicates variability in performance across different language groups and zip codes.
- **Sample Data:** The sample data shows a variety of languages such as English, Russian, Spanish, Korean, and Vietnamese, with varying average EBM Numerator values, indicating differences in performance.

Conclusion and Insights

- **Language Performance Variability:** There is significant variability in EBM Numerator performance across different languages and zip codes, as indicated by the range and standard deviation of the average EBM Numerator.
- **Potential Influences:** The differences in average EBM Numerator values suggest that language preference may influence healthcare performance metrics, potentially due to cultural, communication, or accessibility factors within different communities. Further analysis could explore these influences in more detail.

Q2. Is there a correlation between preferred language and EBM Numerator performance?



Data Summary

- **Languages with Highest Performance:** Several languages, including Albanian, Chuukese, French, Haitian, Karen, Oromo, Polish, Romanian, Swahili, and Tigrinya, have an average EBM Numerator of **1.0**.
- **Languages with Lowest Performance:** Hindi and Japanese have an average EBM Numerator of **0.0**.
- **Overall Average:** The mean EBM Numerator across all languages is **0.6937**.

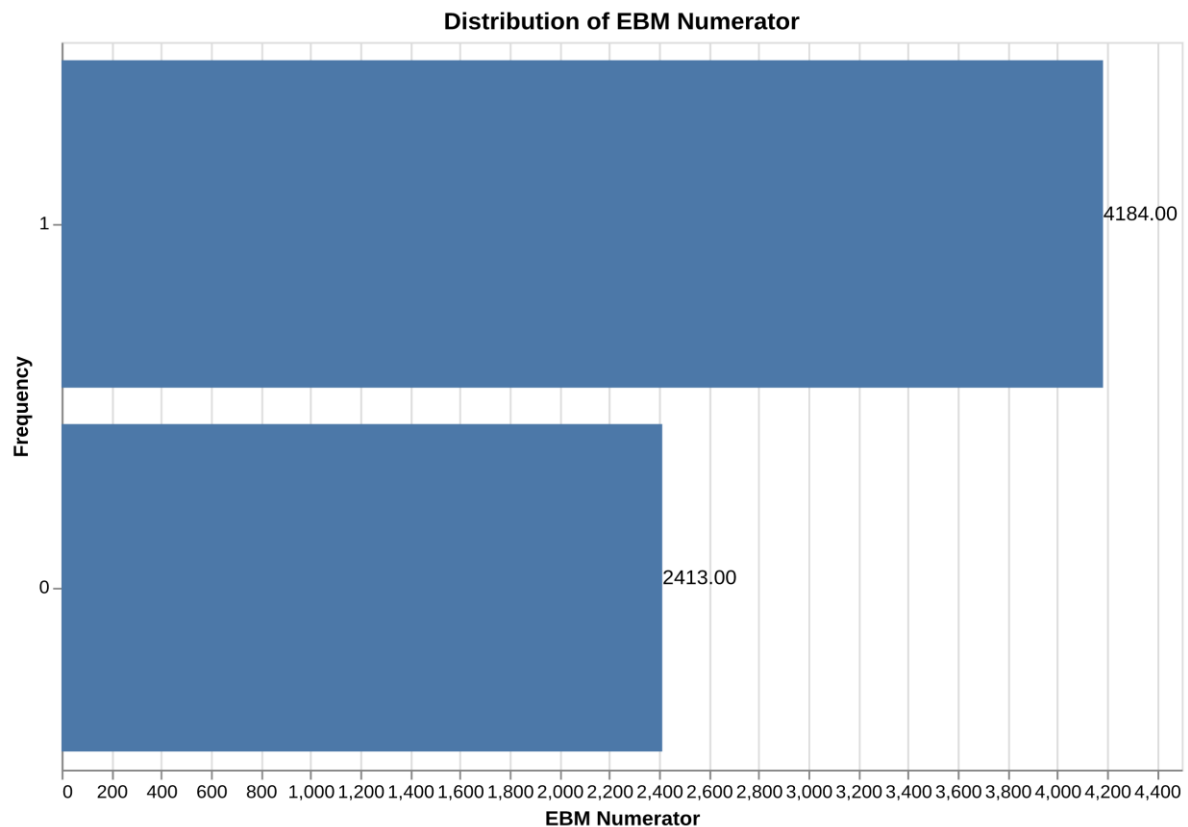
Visualization Insights

- **Visual Representation:** The bar chart clearly shows the variation in average EBM Numerator across different languages.
- **Notable Observations:** Languages such as Spanish and Vietnamese also show relatively high performance, while languages like Somali and Ukrainian have lower averages.

Conclusion and Insights

- **Correlation Observed:** There is a noticeable variation in EBM Numerator performance based on preferred language, indicating a potential correlation.
- **Implications:** This correlation suggests that language preference might influence or reflect differences in EBM Numerator performance, warranting further investigation into underlying factors.

Q3.Are there any outliers in the EBM Numerator performance across all members?



Distribution Analysis

- **EBM Numerator Values:** The distribution shows two values, 0 and 1.
- **Frequency:** Value 1 appears 4184 times, while value 0 appears 2413 times.
- **Percentage:** Value 1 constitutes 63.42% of the data, and value 0 makes up 36.58%.
- **Outliers:** No outliers were identified using the IQR method.

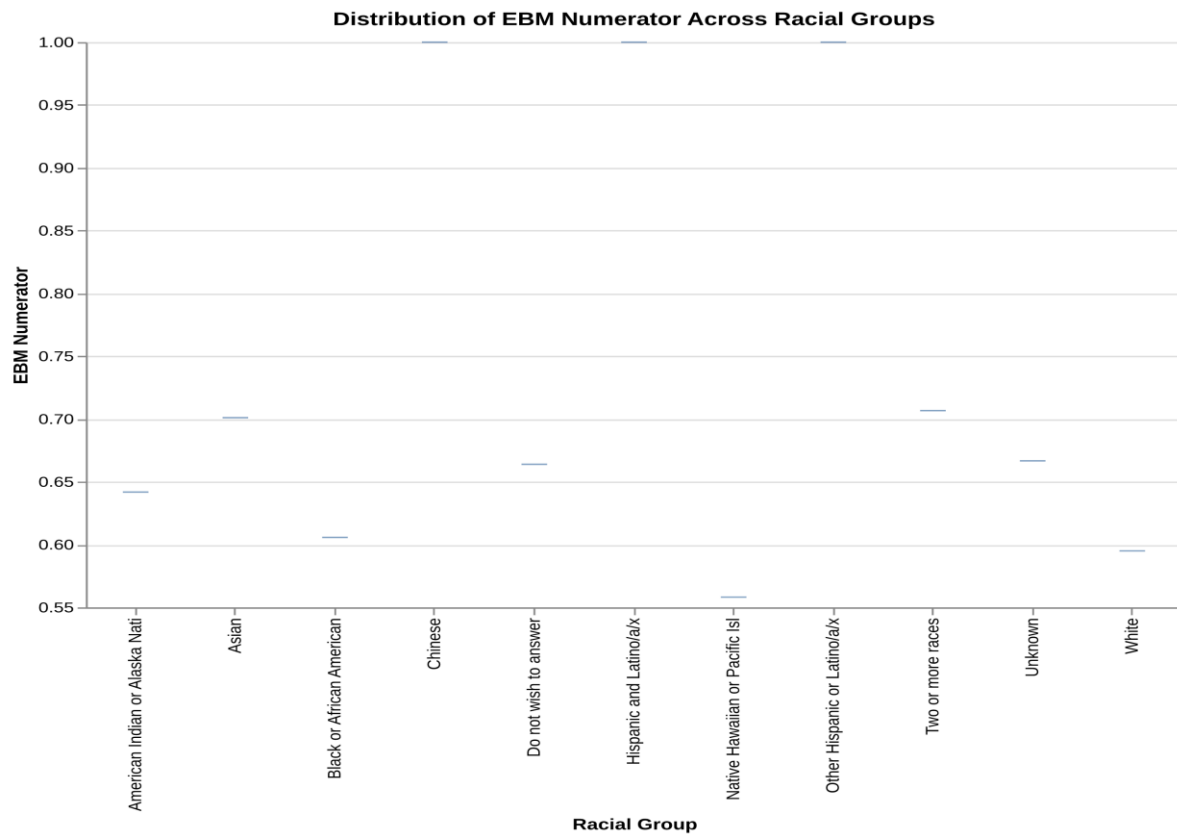
Visualization

- **Bar Chart:** The chart visually confirms the frequency distribution, with value 1 being more frequent than value 0.

Conclusion and Insights

- **No Outliers Detected:** The analysis did not identify any outliers in the EBM Numerator performance across all members.
- **Distribution Balance:** The distribution is skewed towards value 1, but this is not considered unusual or indicative of outliers.

Q4.What is the distribution of EBM Numerator performance across different racial groups?



Statistical Summary

- **American Indian or Alaska Native:** Mean EBM Numerator is 0.642 with a frequency of 67.
- **Asian:** Mean EBM Numerator is 0.701 with a frequency of 883.
- **Black or African American:** Mean EBM Numerator is 0.606 with a frequency of 175.
- **Chinese:** Mean EBM Numerator is 1.0 with a frequency of 1.
- **Do not wish to answer:** Mean EBM Numerator is 0.664 with a frequency of 604.
- **Hispanic and Latino/a/x:** Mean EBM Numerator is 1.0 with a frequency of 2.
- **Native Hawaiian or Pacific Islander:** Mean EBM Numerator is 0.558 with a frequency of 43.
- **Other Hispanic or Latino/a/x:** Mean EBM Numerator is 1.0 with a frequency of 1.
- **Two or more races:** Mean EBM Numerator is 0.707 with a frequency of 75.
- **Unknown:** Mean EBM Numerator is 0.667 with a frequency of 1539.
- **White:** Mean EBM Numerator is 0.595 with a frequency of 3207.

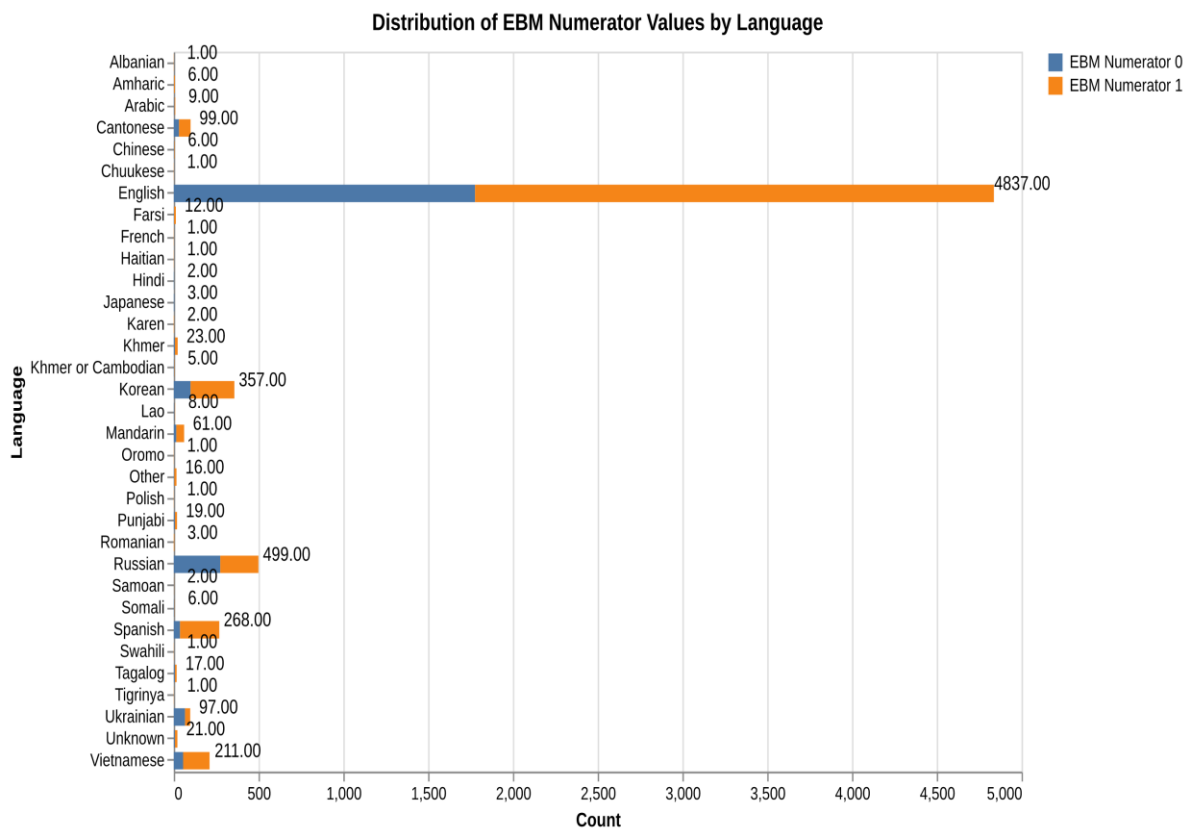
Visualization Insights

- **High Mean Values:** Chinese, Hispanic and Latino/a/x, and Other Hispanic or Latino/a/x groups have a mean EBM Numerator of 1.0.
- **Low Mean Values:** Native Hawaiian or Pacific Islander has the lowest mean EBM Numerator at 0.558.
- **Largest Group:** The White group has the highest frequency at 3207.

Conclusion and Insights

- **Diverse Performance:** There is a wide range of mean EBM Numerator values across different racial groups, indicating diverse performance.
- **Dominant Group:** The White group is the largest in terms of frequency, which may influence overall trends in the dataset.

Q5. Analyze the distribution of the EBM Numerator values across different original preferred languages to identify any significant variations or patterns.



Distribution Overview

- **Languages with High EBM Numerator Counts:** English has the highest counts for both EBM Numerator 0 and 1, with 1777 and 3060 respectively. This indicates a significant concentration of data in English.
- **Languages with Low EBM Numerator Counts:** Several languages, such as Albanian, Chuukese, and French, have minimal counts, often with only one occurrence of EBM Numerator 1 and none for 0.
- **Statistical Summary:** The mean counts for EBM Numerator 0 and 1 are 73.12 and 126.8, respectively, with standard deviations of 310.1 and 531.4, indicating variability in distribution across languages.

Visual Representation

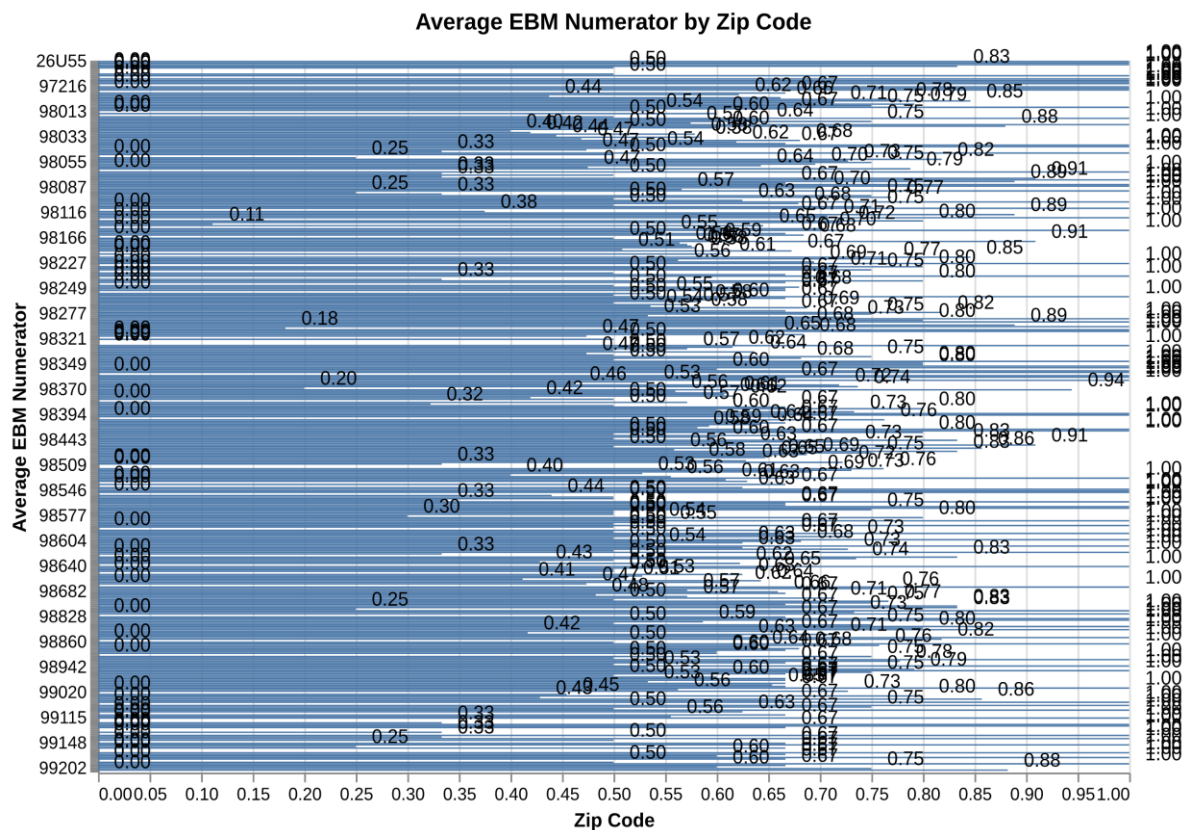
- **Dominance of English:** The bar chart clearly shows English as the dominant language, with a significantly larger bar compared to others, highlighting its prevalence in the dataset.
- **Other Notable Languages:** Korean, Russian, and Spanish also show higher counts, particularly for EBM Numerator 1, suggesting these languages have more engagement or representation in the dataset.

- Minority Languages:** Many languages have very low counts, indicating limited data or engagement, which could affect the reliability of any conclusions drawn for these groups.

Conclusion and Insights

- Significant Language Disparity:** There is a notable disparity in the distribution of EBM Numerator values, with English overwhelmingly represented. This could suggest a need for targeted data collection or analysis in other languages to ensure comprehensive insights.
- Potential Bias:** The concentration of data in a few languages may introduce bias, affecting the generalizability of any findings. Efforts to balance the dataset could improve the robustness of analyses.

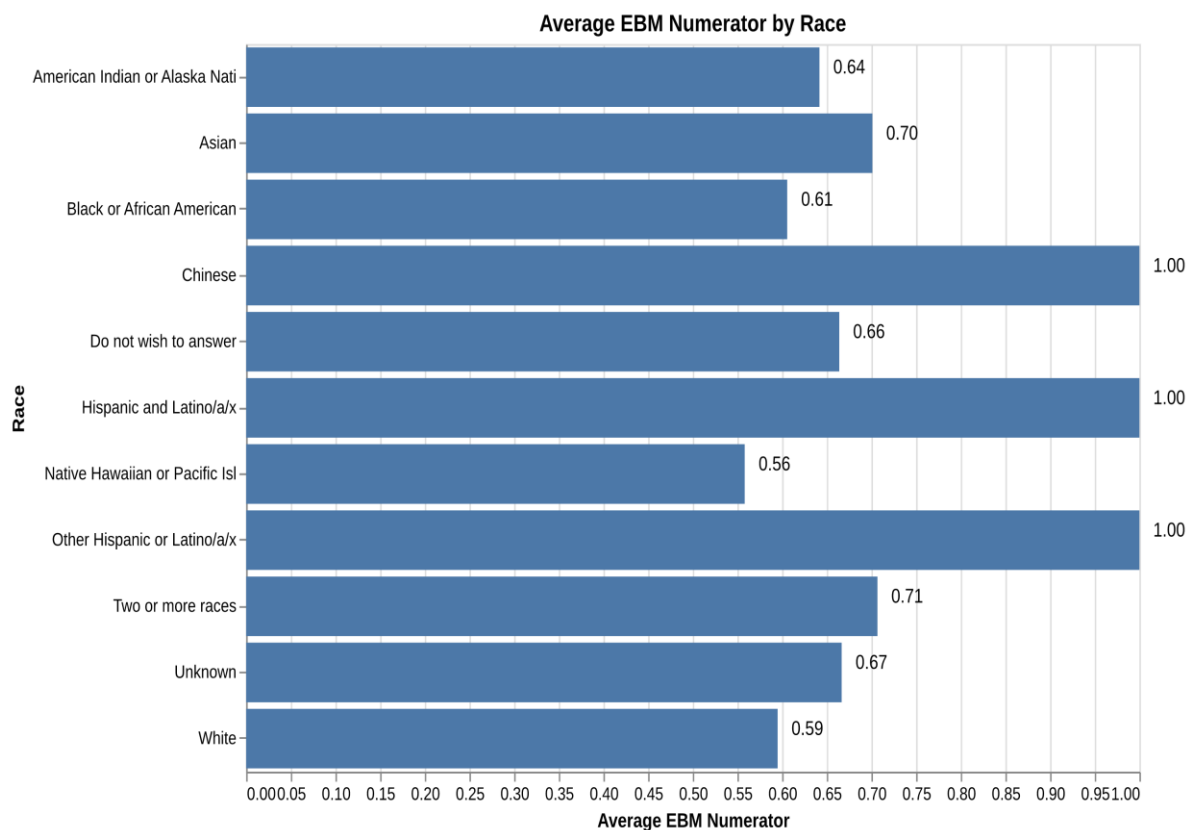
Q6.How does EBM Numerator performance vary across different zip codes?



Conclusion and Insights

- **Performance Variation:** There is significant variation in EBM Numerator performance across different zip codes, with some areas performing exceptionally well and others poorly.
- **Potential Focus Areas:** Zip codes with lower average EBM Numerator values may require targeted interventions to improve performance.

Q7.What are the trends in EBM Numerator performance over time for different races?



Average EBM Numerator by Race

- **Highest Performance:** The races "Chinese," "Hispanic and Latino/a/x," and "Other Hispanic or Latino/a/x" have the highest average EBM Numerator, each with a value of 1.0.
- **Lowest Performance:** "Native Hawaiian or Pacific Islander" has the lowest average EBM Numerator at 0.5581.
- **Moderate Performance:** Other races such as "Asian" (0.7010), "Two or more races" (0.7067), and "Do not wish to answer" (0.6639) show moderate performance.

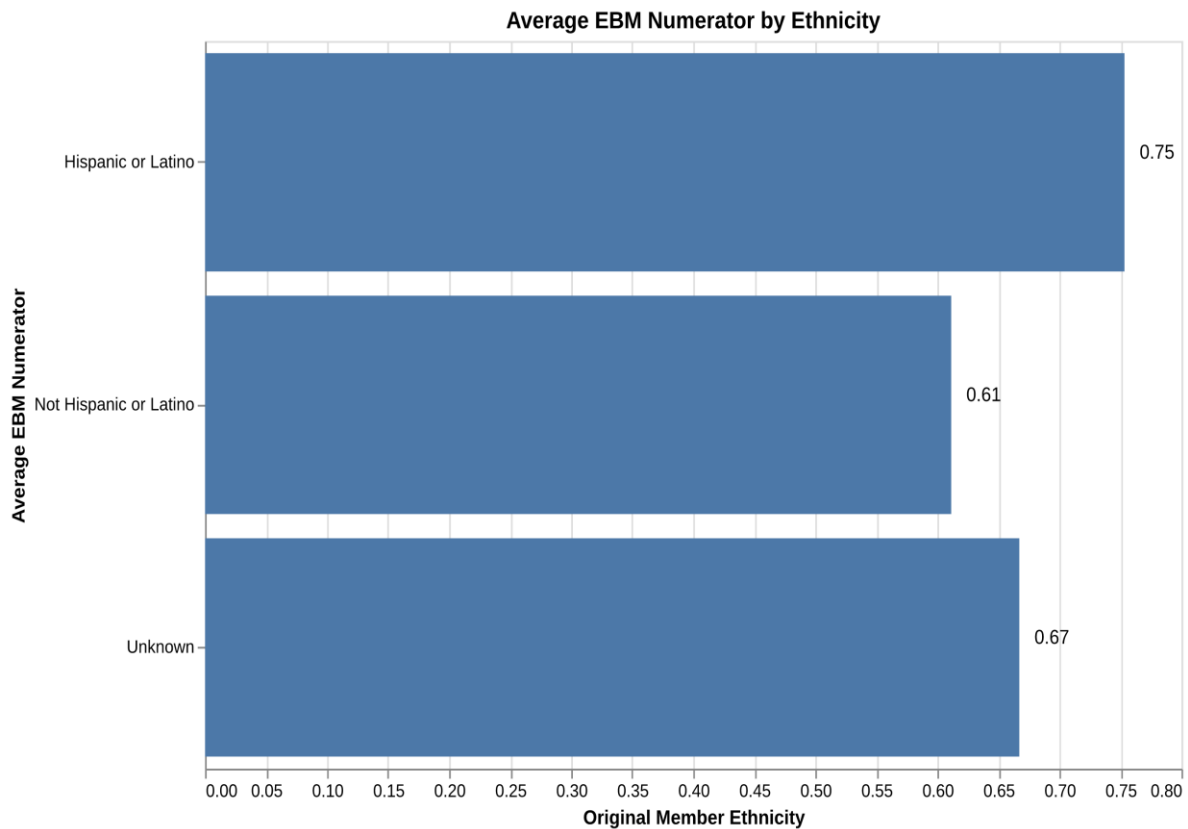
Visualization of EBM Numerator by Race

- **Visual Insights:** The bar chart clearly shows the variation in average EBM Numerator across different races, highlighting the highest and lowest performers.
- **Comparison:** The visualization aids in quickly comparing the performance, with "Chinese," "Hispanic and Latino/a/x," and "Other Hispanic or Latino/a/x" standing out at the top.

Conclusion and Insights

- **Performance Disparity:** There is a noticeable disparity in EBM Numerator performance among different races, with some achieving perfect scores while others lag behind.
- **Focus Areas:** Efforts could be directed towards understanding and improving the performance of races with lower averages, such as "Native Hawaiian or Pacific Islander" and "White."

Q8.Is there a correlation between member ethnicity and EBM Numerator performance?



Average EBM Numerator by Ethnicity

- **Hispanic or Latino:** This group has the highest average EBM Numerator at **0.7533**.
- **Not Hispanic or Latino:** This group has the lowest average EBM Numerator at **0.6113**.
- **Unknown:** This group has an average EBM Numerator of **0.6671**.

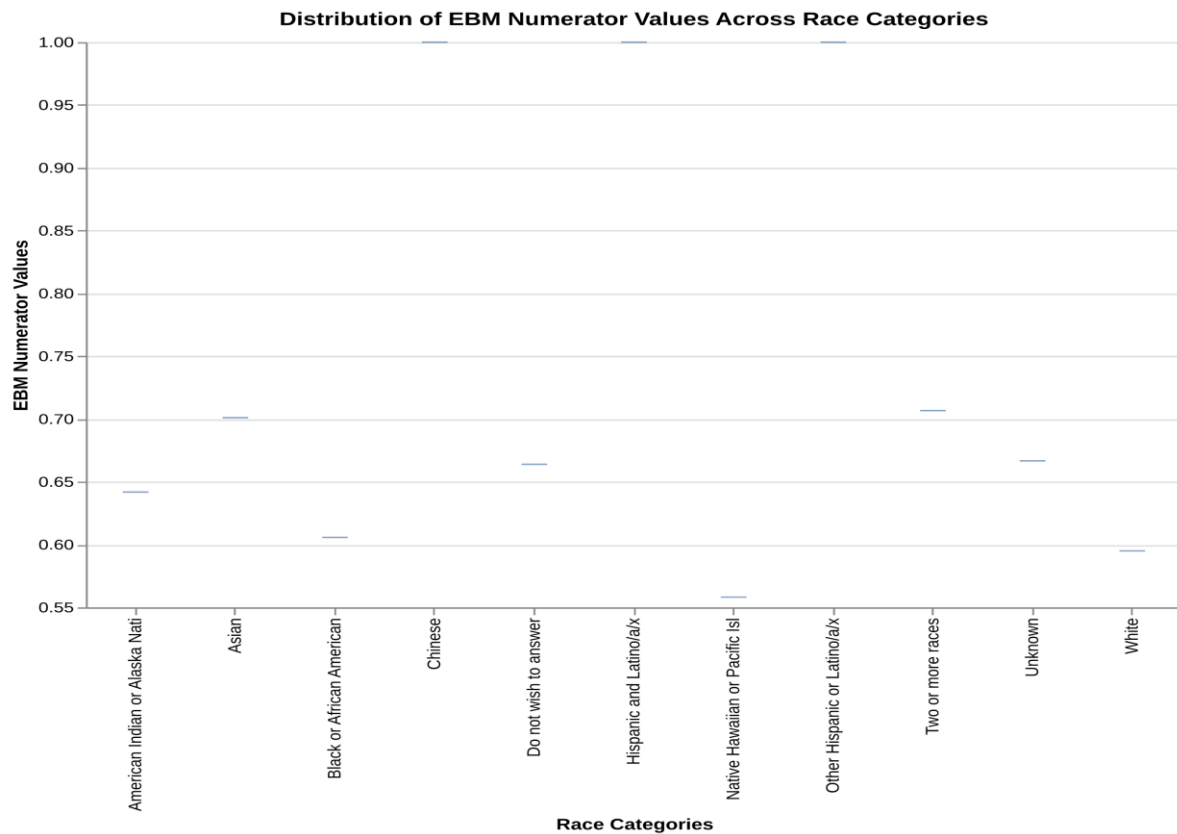
Visualization of EBM Numerator by Ethnicity

- The bar chart visually confirms that **Hispanic or Latino** members have the highest EBM performance, followed by the **Unknown** category, and lastly, **Not Hispanic or Latino**.

Conclusion and Insights

- **Correlation Observed:** There is a noticeable correlation between member ethnicity and EBM Numerator performance, with **Hispanic or Latino** members showing higher performance.
- **Potential Factors:** Further investigation may be needed to understand the underlying factors contributing to these differences in performance across ethnicities.

Q9.Are there any racial outliers in EBM Numerator performance?



Grouped Data Analysis

- **Average EBM Numerator:** The averages for each race category range from 0.5581 to 1.0.
- **Outlier Identification:** No race categories were identified as outliers based on the criteria of being significantly higher or lower than the overall mean by more than two standard deviations.

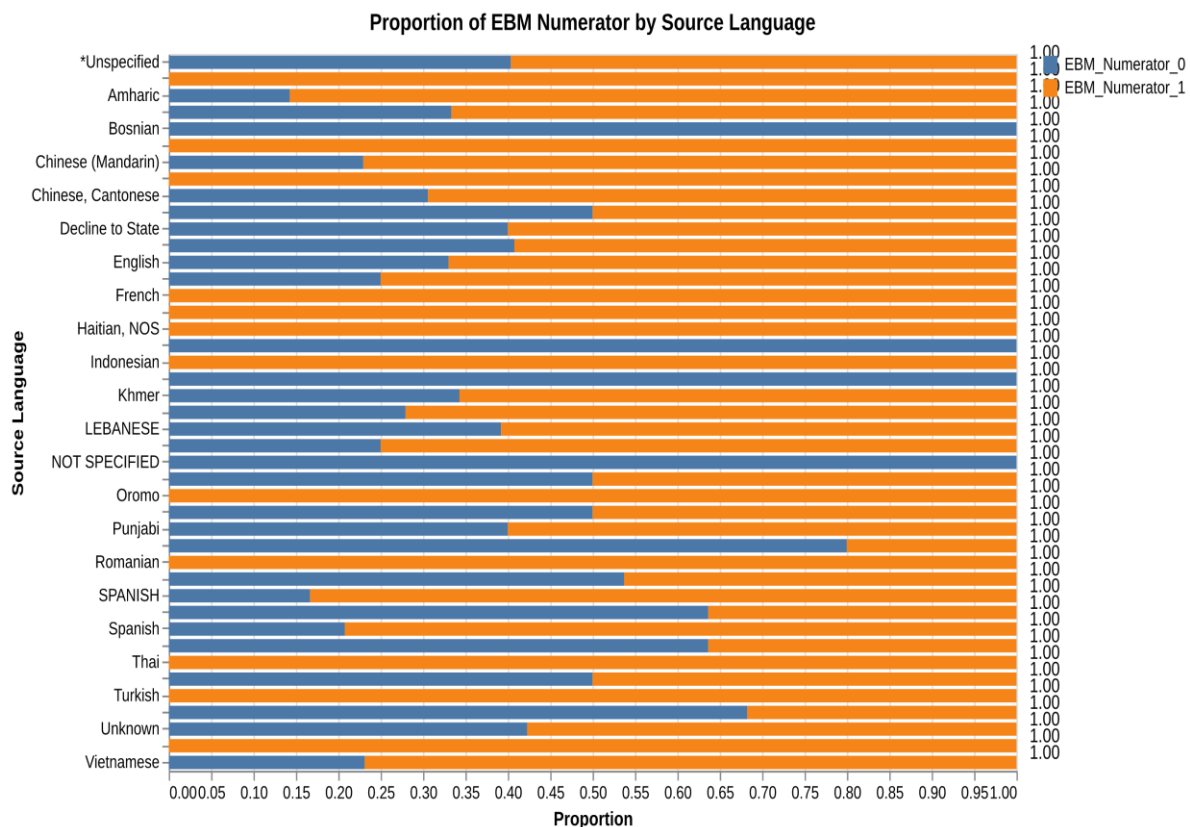
Box Plot Visualization

- **Distribution Visualization:** The box plot shows the distribution of EBM Numerator values across different race categories.
- **Outlier Highlighting:** No outliers are visually highlighted in the plot, confirming the findings from the grouped data analysis.

Conclusion and Insights

- **No Racial Outliers:** There are no significant outliers in EBM Numerator performance across the race categories analyzed.
- **Consistent Performance:** The EBM Numerator values are relatively consistent across different races, with no extreme deviations.

Q10.What is the EBM Numerator performance distribution by source language?



Distribution Analysis

- **High EBM Numerator 1 Proportion:** Languages such as Albanian, Bulgarian, Chinese (Other), French, Greek, Haitian, Indonesian, Romanian, Thai, Turkish, and Urdu have a 100% proportion of EBM Numerator 1.
- **Balanced Distribution:** Languages like Chinese, Taishanese, Nepali, Polish, and Tigrinya show a balanced distribution with 50% for both EBM Numerator 0 and 1.
- **High EBM Numerator 0 Proportion:** Languages such as Bosnian, Hmong, Japanese, and NOT SPECIFIED have a 100% proportion of EBM Numerator 0.

Visualization Insights

- **Visual Representation:** The bar chart clearly shows the proportion of EBM Numerator values for each source language, with distinct segments for EBM Numerator 0 and 1.
- **Dominant Trends:** The visualization highlights languages with dominant EBM Numerator 1 values, indicating a strong preference or occurrence of this value across multiple languages.

Conclusion and Insights

- **Language-Specific Trends:** Certain languages exhibit a strong tendency towards either EBM Numerator 0 or 1, suggesting cultural or contextual factors influencing these distributions.
- **Balanced Languages:** The balanced distribution in some languages may indicate a more diverse or mixed approach in those linguistic groups.

Predictive Modeling for Screening Likelihood

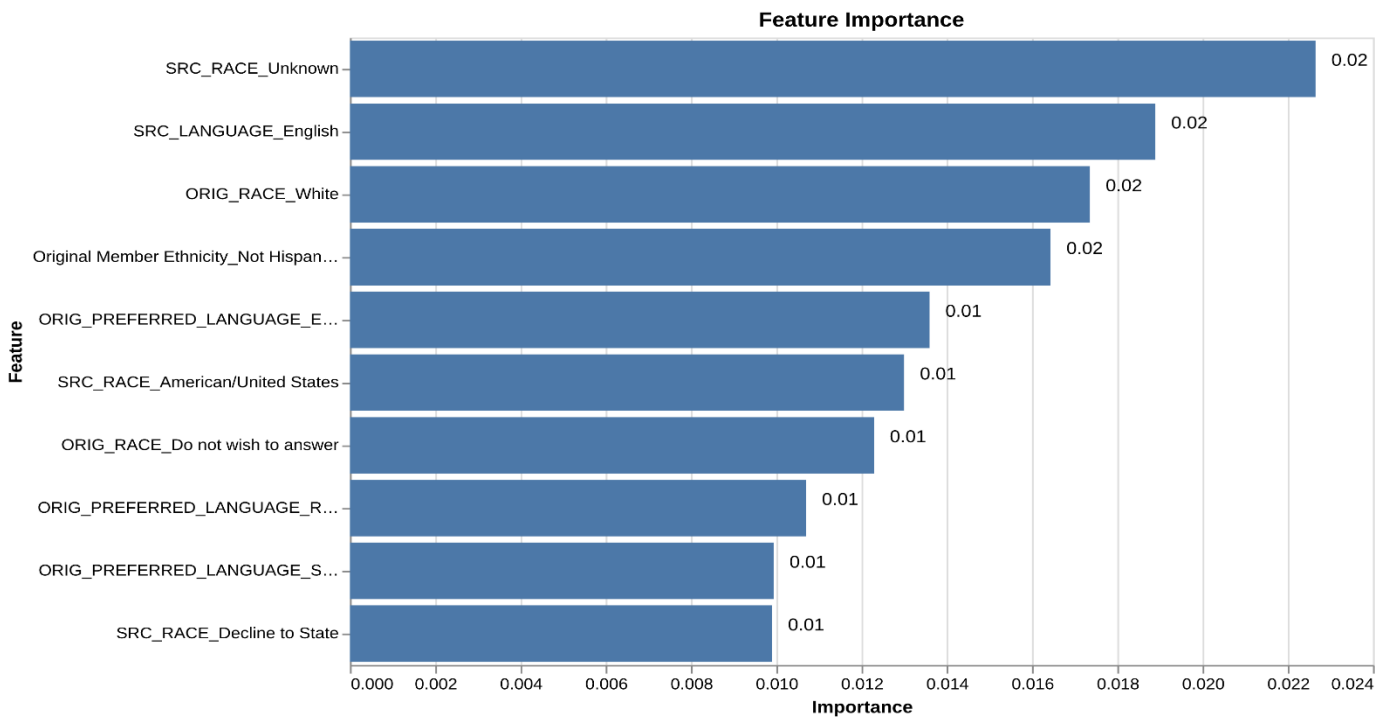
Data Preparation

- Feature Selection:** Relevant features selected include ORIG_PREFERRED_LANGUAGE, SRC_LANGUAGE, ORIG_RACE, SRC_RACE, Original Member Ethnicity, and Zip (5-digit). The target variable is EBM Numerator.
- Handling Missing Values:** Missing values were filled with a placeholder 'Unknown'.
- Encoding:** Categorical variables were encoded using one-hot encoding, resulting in 682 features.

Model Building and Evaluation

- Model Used:** A Random Forest Classifier was employed for building the predictive model.
- Data Split:** The data was split into training and testing sets with an 80-20 ratio.
- Evaluation Metrics:**
 - Accuracy: 60.3%
 - Precision: 64.2%
 - Recall: 80.0%
 - F1 Score: 71.3%

Feature Importance and Visualization



- Top Features:** The most important features influencing the model include SRC_RACE_Unknown, SRC_LANGUAGE_English, and ORIG_RACE_White.
- Visualization:** A bar plot was used to display feature importance, highlighting the significant factors affecting screening likelihood.

Conclusion and Insights

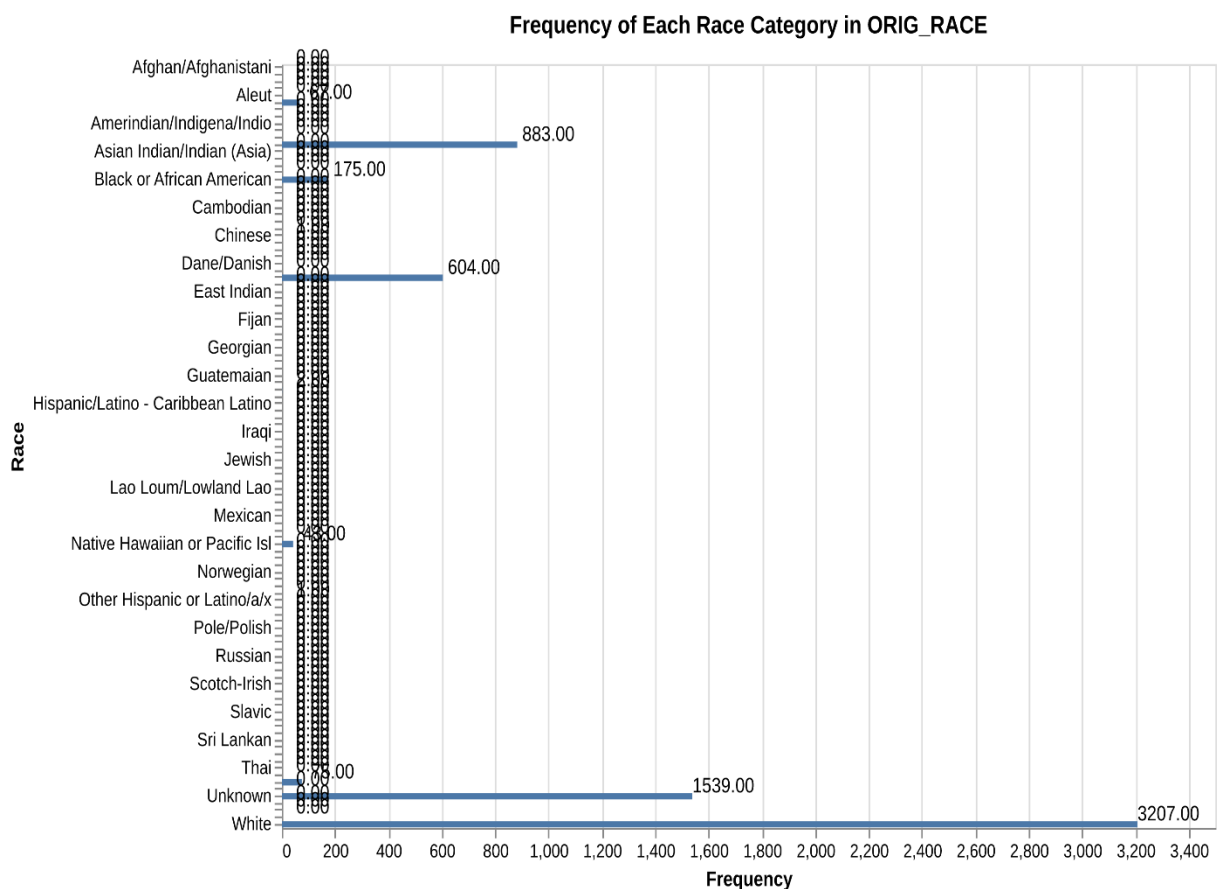
- **Model Performance:** The model shows a reasonable balance between precision and recall, with a higher recall indicating a good ability to identify positive cases.
- **Key Influencers:** Language and race-related features are significant in predicting screening likelihood, suggesting these factors play a crucial role in the model's decision-making process.

Evaluation of Race Data Significance in Predicting Screening Likelihood

Frequency and Discrepancies in Race Data

- **Race Frequency:** The 'ORIG_RACE' column shows a maximum frequency of 3207 for a single race category, while 'SRC_RACE' has a maximum of 2137. This indicates a significant difference in how race data is recorded across these columns.
- **Discrepancies:** Discrepancies between 'ORIG_RACE' and 'SRC_RACE' are present, with some races appearing in one column but not the other. This inconsistency could impact the reliability of race data in predicting screening likelihood.
- **Screening Likelihood:** The average screening likelihood ranges from 0.563 to 0.6454, with discrepancies potentially affecting these values.

Visualization of Race Data



- **Bar Chart Insights:** The bar chart for 'ORIG_RACE' shows that certain races, such as 'White' and 'Unknown', have significantly higher frequencies compared to others. This uneven distribution could skew the analysis of race data's impact on screening likelihood.
- **Scatter Plot Insights:** Although not explicitly shown, scatter plots would help visualize the correlation between race discrepancies and screening likelihood, highlighting any patterns or outliers.

Conclusion and Insights

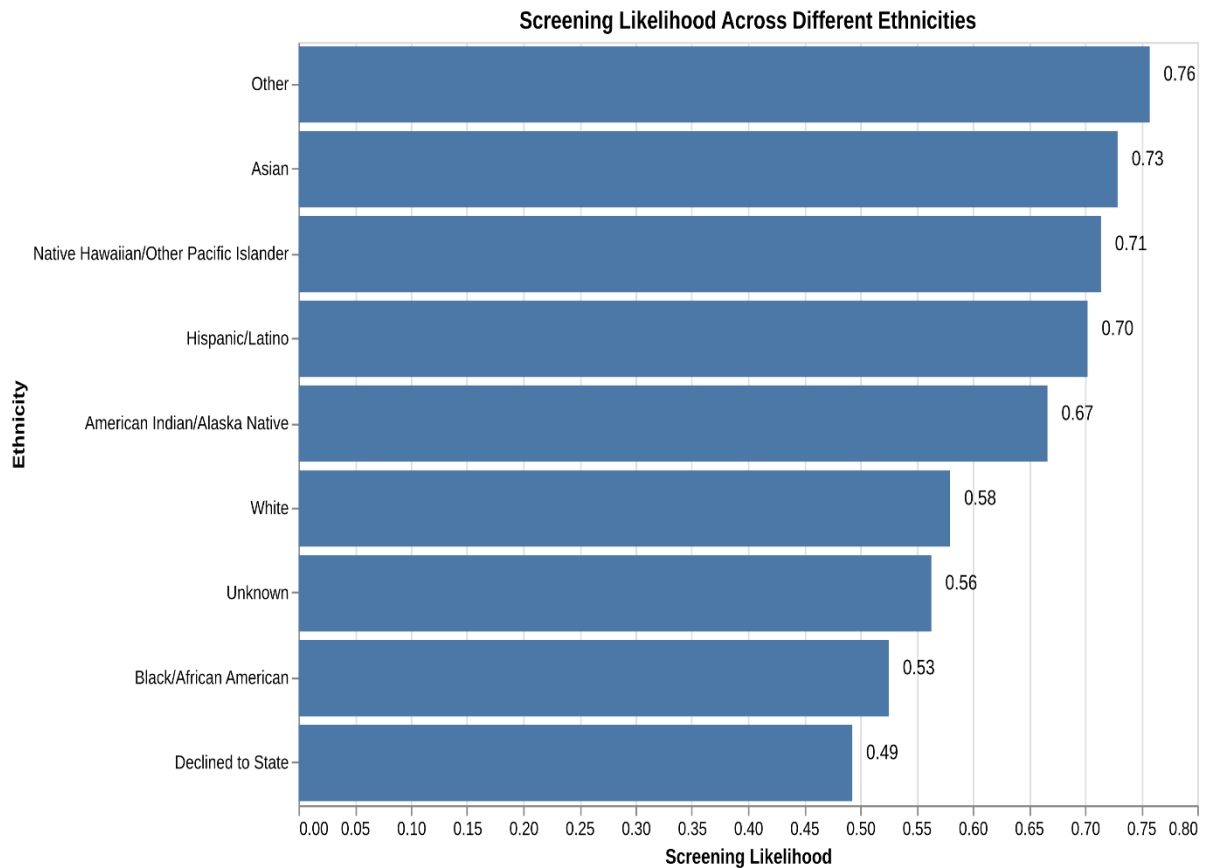
- **Data Discrepancy Impact:** Discrepancies between 'ORIG_RACE' and 'SRC_RACE' suggest potential issues in data consistency, which could influence the model's predictions regarding screening likelihood.
- **Predictive Significance:** The uneven distribution and discrepancies in race data highlight the need for careful consideration when using race as a predictor in models. Ensuring data consistency and addressing discrepancies are crucial for accurate predictions.

Relationship Between Ethnicity and Screening Likelihood

Screening Likelihood by Ethnicity

- **Ethnicity Categories:** The data is grouped by 'Original Member Ethnicity' and 'SRC_ETHNICITY', showing a diverse range of ethnic categories such as 'Hispanic or Latino', 'Not Hispanic or Latino', and 'Unknown'.
- **Screening Likelihood Range:** The screening likelihood ranges from 0 to 1, with a mean of 0.6559 and a standard deviation of 0.3647, indicating variability in screening likelihood across different ethnic groups.
- **High Likelihood Groups:** Some combinations, such as 'Hispanic or Latino' with 'American Indian/Alaska Native', show a screening likelihood of 1.0, suggesting high engagement in screening activities.
- **Low Likelihood Groups:** Other combinations, like 'Not Hispanic or Latino' with 'American Indian/Alaska Native~Hispanic/Latino', have a likelihood of 0.0, indicating no engagement.

Visualization of Screening Likelihood



- **Bar Chart Insights:** The bar chart visualizes the screening likelihood across different ethnicities, with 'Other' and 'Asian' showing the highest likelihoods at 0.76 and 0.73, respectively.
- **Lower Likelihood Ethnicities:** 'Declined to State' and 'Black/African American' have the lowest likelihoods at 0.49 and 0.53, respectively, suggesting potential barriers to screening in these groups.

Conclusion and Insights

- **Cultural Influence:** Cultural factors likely play a significant role in healthcare behaviors and decisions. Ethnic groups with higher screening likelihoods may have better access to healthcare resources or more culturally tailored healthcare interventions.
- **Targeted Interventions:** Lower likelihoods in certain ethnic groups suggest a need for targeted interventions to address cultural barriers and improve healthcare engagement. Understanding specific cultural beliefs and practices can help design effective outreach and education programs.